ModelIt!

Teacher's Guide

Where Does All the Plastic Go?

Modeling Plastic Pollution Pathways, Microplastic Accumulation, and Bioaccumulation

Quick Reference

Grade Level	Grade 12L2
NGSS Standard	HS-ESS3-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-4

Evaluate or refine a technological solution that reduces impacts of human activities on natural systems.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze data from their plastic pollution models to evaluate the effectiveness of different intervention strategies and identify the most impactful leverage points in the plastic lifecycle.
Disciplinary Core Idea	**ESS3.C Human Impacts on Earth Systems** — Human activities have significantly altered the biosphere through the production and disposal of persistent synthetic materials, and the consequences of plastic pollution extend through all Earth systems including the hydrosphere, atmosphere, lithosphere, and biosphere.
Crosscutting Concept	**Cause and Effect** — Students identify the causal chain from plastic production through environmental accumulation, fragmentation, bioaccumulation, and human exposure, and evaluate how interventions at different points in this chain produce different outcomes.

CAST Testing Alignment

MULTIPLE CHOICE:

A community triples its recycling rate from 8% to 24% while plastic production in the region continues growing at 4% per year. According to the model, what is the most likely outcome after 20 years? A) Environmental plastic load decreases significantly B) Environmental plastic load continues growing but more slowly C) Microplastic concentration decreases D) The community achieves zero waste

CONSTRUCTED RESPONSE:

Using your plastic pollution model, explain why microplastic concentrations in top predators like tuna fish are thousands of times higher than microplastic concentrations in the surrounding ocean water. Reference the concepts of bioaccumulation and biomagnification and at least three model components in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Plastic is arguably the most successful and most destructive material humanity has ever created. Since mass production began in the 1950s, we have manufactured 8.3 billion metric tons of plastic — more than one ton for every person alive today. Plastic\'s extraordinary durability, which makes it so useful, is also what makes it so dangerous: unlike natural materials, plastic does not biodegrade. It fragments into smaller and smaller particles that persist for centuries, spreading through water, air, and soil to reach every corner of the planet and every level of the food web.

The Scale of the Problem

Global plastic production has increased from 2 million tons in 1950 to over 400 million tons in 2024, growing at approximately 4% per year. Of the 8.3 billion metric tons produced historically, approximately 6.3 billion tons are now waste. Only 9% has been recycled, 12% incinerated, and 79% accumulated in landfills or the natural environment. An estimated 8-12 million metric tons of plastic enter the ocean every year — equivalent to dumping a garbage truck of plastic into the ocean every minute. At current rates, plastic in the ocean will outweigh fish by 2050. This is not a waste management problem that better recycling can solve — it is a production problem driven by a petrochemical industry that plans to triple plastic output by 2060.

From Macro to Micro: The Fragmentation Cascade

Large plastic items exposed to UV radiation, wave action, temperature fluctuations, and biological activity gradually fragment into smaller pieces. A single plastic bottle can break into thousands of microplastic fragments (less than 5mm) and eventually into nanoplastics (less than 1 micrometer) that are invisible to the naked eye. This process takes decades to centuries and is irreversible — once a microplastic exists, it persists in the environment indefinitely. Critically, fragmentation increases the total surface area of plastic, which increases its capacity to absorb toxic chemicals from the surrounding environment. Microplastics act as tiny sponges that concentrate persistent organic pollutants to levels 100,000 to 1,000,000 times higher than surrounding water.

Bioaccumulation: Plastic in the Food Web

Marine organisms encounter microplastics throughout the food web. Zooplankton and filter feeders like mussels and oysters ingest microplastics directly from the water. Small fish eat contaminated zooplankton. Larger fish eat contaminated small fish. At each trophic level, plastic and its associated toxins accumulate — a process called biomagnification. Studies have found microplastics in 100% of marine turtle species, 59% of whale species, and 40% of seabird species examined. Humans are not exempt: recent studies have detected microplastics in human blood (80% of samples), human lungs, human placenta, and breast milk. The health effects are still being researched but include potential endocrine disruption, inflammation, and cellular damage.

Solutions: From Cleanup to System Change

Current solutions span a spectrum from downstream cleanup to upstream system change. Ocean cleanup projects remove floating plastic but cannot address the vast majority that sinks or fragments into microplastics. Improved recycling helps but is fundamentally limited by economics — virgin plastic from cheap oil is often cheaper than recycled material. The most effective interventions target the source: banning single-use plastics (bags, straws, packaging), implementing Extended Producer Responsibility that makes manufacturers pay for waste management, developing truly biodegradable alternatives, and fundamentally redesigning products for circular economy principles. The Global Plastics Treaty negotiations represent the most significant international effort, aiming to reduce plastic production and create binding regulations — but face opposition from the petrochemical industry that profits from continued growth.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Plastic Production Rate, Waste Management Capacity, Environmental Plastic Load, Microplastic Concentration, Bioaccumulation in Food Webs, Marine Ecosystem Health, and Human Health Exposure as key components of the plastic pollution system.

E **Establish Relationships**

Students determine that production rate overwhelms waste management capacity, that environmental plastic fragments into persistent microplastics over time, that bioaccumulation concentrates plastics up food chains to humans, and that only source reduction meaningfully reduces accumulation.

V **Visualize & Model**

Students build a computational model showing how plastic flows from production through environmental pathways to biological accumulation, revealing why downstream solutions alone are insufficient.

E **Evaluate Outcomes**

Students run business-as-usual, improved-recycling, and system-transformation scenarios to test predictions about which interventions can reverse the growing plastic pollution trend.

R **Revise & Extend**

Students add chemical leaching, atmospheric transport, or extended producer responsibility to explore how additional factors affect pollution pathways and the effectiveness of policy interventions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"8.3 billion metric tons of plastic. Where did it all go? Today we\'re tracing every gram of it through the environment and into our own bodies."*

Watch For

Show cover image. Shock stat: Only 9% of all plastic ever made has been recycled. Ask: Where do you think the other 91% went?

Slide 2: Learning Objectives

3 min

Say: *"We\'re not just learning about trash today — we\'re modeling a global system that connects your plastic water bottle to microplastics in your bloodstream."*

Watch For

Have students read objectives. Pre-teach vocabulary. Key visual: show the size comparison of microplastics — smaller than a grain of rice, found everywhere on Earth.

Slide 3: Big Question

3 min

Say: *"Plastic has been found in human blood, lungs, placenta, and breast milk. How did it get there? Trace the pathway from a plastic bag to your body."*

Watch For

Quick-write: Students write their initial hypothesis about how plastic reaches the human body. Save for comparison after modeling.

Slide 4: LEVER Framework

2 min

Say: *"We\'ll model the entire lifecycle of plastic — from production to your bloodstream. This model will reveal why some solutions work and others don\'t."*

Watch For

Preview LEVER steps. Emphasize that this model will challenge assumptions about recycling as a solution. The model follows the plastic, not the narrative.

Slide 5: Activity 1: Components

8 min

Say: *"Which components are external drivers of the plastic pollution system, and which are internal consequences? Think about what pushes plastic into the environment versus what happens once it\'s there."*

Watch For

Guide sorting: External = Plastic Production Rate, Waste Management Capacity (human decisions). Internal = the five environmental and biological responses. Discuss: Is human health exposure an endpoint or a feedback that could change production policy?

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the journey: production to environment to microplastics to food web to your body. Where are the positive feedback loops? Where are the bottlenecks that determine how much plastic reaches each stage?"*

Watch For

Students draw arrows with + or - signs. Key insight: the one-way nature of plastic fragmentation — once microplastics exist, they persist essentially forever. There is no natural cleanup mechanism.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run all three scenarios. Pay attention to the recycling-only scenario — does tripling recycling solve the problem? Why or why not?"*

Watch For

Students run scenarios and compare outcomes. Key discussion: Why does improved recycling barely slow the trend? Guide them to see that 4% production growth overwhelms even tripled recycling. Only source reduction changes the trajectory.

Slide 8: Discoveries

5 min

Say: *"The biggest finding: recycling is necessary but mathematically insufficient. When production grows faster than recycling improves, the problem gets worse no matter how much you recycle."*

Watch For

Class discussion. Compare initial hypotheses to model evidence. Challenge question: If recycling alone can\'t solve this, what does that tell us about where the real solution must come from?

Slide 9: STEM Challenge

12 min

Say: *"Your community generates 15,000 tons of plastic waste per year and it\'s contaminating local fisheries. Design a system that eliminates plastic pollution at every stage — from what enters the community to what leaves it."*

Watch For

Groups design comprehensive systems addressing production reduction, material substitution, waste management, and environmental remediation. Present with evidence from model simulations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Plastic Production Rate, Waste Management Capacity
- **INTERNAL:** Environmental Plastic Load, Microplastic Concentration, Bioaccumulation in Food Webs, Marine Ecosystem Health, Human Health Exposure

Relationships

- **Plastic Production Rate to Environmental Plastic Load:** POSITIVE (+) — Higher production rates generate more plastic waste, and since waste management captures only a fraction, the excess accumulates in the environment. A 4% annual production increase outpaces improvements in recycling.
- **Waste Management Capacity to Environmental Plastic Load:** NEGATIVE (-) — Better waste management — higher recycling rates, improved collection, safer disposal — reduces the fraction of produced plastic that leaks into the environment, though it cannot offset unlimited production growth.
- **Environmental Plastic Load to Microplastic Concentration:** POSITIVE (+) — More plastic in the environment means more material undergoing fragmentation through UV exposure, physical abrasion, and biological processes, steadily increasing the concentration of microplastic particles.
- **Microplastic Concentration to Bioaccumulation in Food Webs:** POSITIVE (+) — Higher environmental microplastic concentrations mean more particles are ingested by organisms at the base of the food web, which then concentrate through biomagnification at each trophic level.
- **Bioaccumulation in Food Webs to Human Health Exposure:** POSITIVE (+) — As microplastics and their associated toxins accumulate in seafood and other food products, human dietary exposure increases proportionally, particularly for populations that consume large amounts of fish and shellfish.
- **Marine Ecosystem Health to Bioaccumulation in Food Webs:** NEGATIVE (-) — Declining marine ecosystem health reduces the abundance and diversity of organisms, but the surviving organisms in degraded ecosystems often have higher plastic loads because there are fewer organisms to dilute the plastic across the food web.

Key Discoveries

- Improved recycling alone cannot solve the plastic crisis because production growth outpaces recycling capacity — even tripling recycling rates only slightly slows the growth of environmental plastic accumulation when production continues increasing at 4% per year
- Microplastic Concentration continues increasing for decades even after new plastic input stops because existing macroplastics in the environment continue fragmenting — the microplastic problem has significant time lag built into it
- Bioaccumulation means that even low environmental concentrations of microplastics result in high tissue concentrations in top predators including humans — the food web acts as a concentration mechanism that amplifies exposure
- Only the system transformation scenario — reducing production, improving recycling, and implementing circular design — produces a meaningful decrease in Environmental Plastic Load, and even then, the microplastic legacy persists for generations

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Recycling solves the plastic problem"	Only 9% of all plastic ever produced has been recycled, and recycling rates have not improved significantly in decades. Many plastics cannot be technically recycled, contamination reduces quality, and recycled plastic is often more expensive than virgin material made from cheap oil. Furthermore, most recycling is actually downcycling — converting a bottle into a lower-quality product that is eventually landfilled anyway. The model shows that even dramatically increased recycling cannot offset continued production growth.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Biodegradable plastics solve the problem"	Most products labeled biodegradable only break down under specific industrial composting conditions (high temperature, specific microorganisms) that don\'t exist in landfills or oceans. In landfills, biodegradable plastics behave identically to conventional plastics for decades. In the ocean, they fragment into microplastics just like conventional plastic. Truly compostable plastics can be part of the solution but only with proper collection infrastructure and honest labeling — otherwise they give consumers a false sense of environmental safety.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Ocean cleanup can remove the plastic that\'s already there"	Ocean cleanup projects like The Ocean Cleanup can remove some floating macroplastics, but they cannot capture the vast majority of ocean plastic — 94% of which has sunk below the surface or fragmented into microplastics too small to collect. The surface plastic that cleanup projects target represents a small fraction of total ocean plastic. Furthermore, cleanup addresses the symptom while the cause — continued production and dumping — adds new plastic faster than any cleanup can remove it. The model shows that cleanup without source reduction is like mopping the floor while the faucet overflows.